ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** evaporation	Slow vaporisation.
2	**(b)** evaporation and condensation	Water vapour cools.
3	**(d)** drip irrigation	Saves water.
4	**(a)** boiling	Vaporisation.
5	**(c)** ice – solid	Ice is solid.
6	**(c)** rainwater harvesting	Water seeps into soil.
7	**(d)** water boils	Boiling point.
8	**(b)** heat is used to change the state	Energy is used for melting.
9	**(c)** roots hold soil and water	Roots help soak water.
10	**(c)** slows down	Humid air.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat is used to change water into steam (1); It is not used to raise the temperature (1).	1+1
12	More heat flows in (1); Ice absorbs heat and melts (1).	1+1
13	Evaporation through pores cools it (1); Heat taken away (1).	1+1
14	Water vapour in the air condenses on the cooler mirror (1); Droplets form on the surface (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Evaporation from the sea (1); Condensation into clouds (1); Precipitation and flow back (1).	1+1+1
16	Only water evaporates (1); Salts remain (1); Condensed water is fresh (1).	1+1+1
17	Irrigation (1); Electricity (1); Flood control (1).	1+1+1
	OR: Put a few drops of spirit on the hand (1); It evaporates quickly (1); Hand feels cool (1).	1+1+1
18	Sunny: fast (1); Cloudy: slow (1); Heat increases evaporation (1).	1+1+1
	OR: Kills germs (1); Prevents diseases (1); Safe to drink (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take plates of different size with same water (1); Keep one in sun and one in shade (1); Use a fan for one (1); Note time to dry (1); Large, warm, windy dries fastest (1).	1+1+1+1+1
	OR: Filter through cloth (1); Boil for 10 minutes (1); Cool (1); Store in a clean covered vessel (1); Kills germs (1).	1+1+1+1+1
20	Sun heats the sea (1); Vapour rises (1); Condenses into clouds (1); Droplets grow (1); Fall as rain (1).	1+1+1+1+1
	OR: Rivers flow to sea (1); Sea gives vapour (1); Clouds give rain (1); Rain recharges groundwater (1); Groundwater feeds rivers (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing evaporation with **boiling**: evaporation happens at any temperature from the surface, boiling at about 100 degrees C.
- Mixing condensation (vapour to liquid) with evaporation (liquid to vapour).
- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** heat is used for changing water to steam	Heat used in vaporisation.
2	**(a)** less water evaporates in the cooler evening	Cooler time.
3	**(d)** spread in the Sun on a windy day	Heat, wind, area.
4	**(b)** increase it	Faster.
5	**(a)** condensation of steam	Steam cools.
6	**(d)** air's water vapour	Condensation.
7	**(c)** a hot, dry day	Dry air.
8	**(b)** steam carries extra heat	It condenses on skin releasing heat.
9	**(c)** vapour condenses on the cooler lid	Condensation.
10	**(d)** shape and volume	States depend on particles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ice floats and insulates (1); Water below remains liquid (1).	1+1
12	Grass cools quickly (1); Vapour condenses on it (1).	1+1
13	Higher temperature speeds up evaporation (1); More vapour escapes into the air (1).	1+1
14	Steam is a gas (1); Hot water is a liquid (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Temperature rises steadily (1); Reaches 100 °C boiling (1); Stays constant during boiling (1).	1+1+1
16	Ice: fixed shape and volume (1); Water: fixed volume, takes container shape (1); Steam: no fixed shape or volume (1).	1+1+1
17	Freezing: liquid to solid (1); Condensation: vapour to liquid (1); Both on cooling (1).	1+1+1
	OR: Sun gives heat (1); Evaporation faster (1); Night is cold and damp (1).	1+1+1
18	Level of groundwater (1); Overuse of wells lowers it (1); Less recharge adds to the problem (1).	1+1+1
	OR: Heats the water (1); Drives evaporation (1); Gives rain (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Supplies fresh water (1); Rains for crops (1); Moderates climate (1); Fills rivers (1); Refills groundwater (1).	1+1+1+1+1
	OR: Ice: fixed shape and volume (1); Water: fixed volume, takes the shape of its container (1); Steam: no fixed shape or volume (1); Heat changes ice to water and water to steam (1); It is the same substance throughout (1).	1+1+1+1+1
20	Ice warms to 0 °C (1); Melts at 0 °C (1); Water warms to 100 °C (1); Boils at 100 °C (1); Steam warms above 100 °C (1).	1+1+1+1+1
	OR: Fill glass with ice (1); Wipe it dry outside (1); Droplets form (1); Vapour from air condenses (1); Conclusion (1).	1+1+1+1+1

Mistakes to avoid:

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- Calling the white cloud over a boiling kettle water vapour; vapour is invisible, the white cloud is tiny droplets.
- Saying that drops on a cold glass come from inside the glass; they come from **vapour in the air**.
- Writing that humid air helps drying; more moisture in the air makes evaporation slower.
- Forgetting that ice melts and water freezes at the same temperature, 0 degrees C.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** evaporation, condensation, precipitation	Standard order.
2	**(c)** condensed vapour	Droplets from breath.
3	**(a)** ice	Solid.
4	**(c)** oceans	Oceans hold most.
5	**(a)** evaporation and seeping	Water goes to air and soil.
6	**(c)** water	Liquid.
7	**(b)** less rain and more run-off	Reduced transpiration.
8	**(d)** ice floats and insulates	Insulation.
9	**(b)** steam	Gas.
10	**(c)** dumping waste in rivers	Pollution.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Release water vapour (1); Hold soil and recharge groundwater (1).	1+1
12	Greater surface area (1); More air contact (1).	1+1
13	Reduce evaporation (1); Prevent dirt (1).	1+1
14	Water is wasted (1); Drops add up (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cities have much concrete (1); Less water seeps (1); Harvesting recharges groundwater (1).	1+1+1
16	Water on pads (1); Fan blows air (1); Evaporation cools air (1).	1+1+1
17	Do not dump waste (1); Treat sewage (1); Keep sources clean (1).	1+1+1
	OR: Water evaporates (1); Vapour condenses on cooler walls (1); Cycle in miniature (1).	1+1+1
18	Blowing removes vapour (1); Faster evaporation (1); Evaporation cools (1).	1+1+1
	OR: Heat (1); Evaporation (1); Wind (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fix leaking taps (1); Rain harvesting (1); Reuse RO water (1); Water meter readings (1); Compare monthly (1).	1+1+1+1+1
	OR: Surface vs whole (1); Any temperature vs 100 °C (1); Slow vs fast (1); Bubbles absent vs present (1); Both give vapour (1).	1+1+1+1+1
20	Rain is uneven (1); Hill areas get heavy rain (1); Some plains get less (1); Poor drainage (1); Management needed (1).	1+1+1+1+1
	OR: Roots hold soil (1); Soil absorbs water (1); Trees reduce runoff (1); Recharge groundwater (1); Prevent floods (1).	1+1+1+1+1

Mistakes to avoid:

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